

Synthetic Satellite Segmentation

Plain-language project progress review

The big question

Can carefully created synthetic satellite-style images help a land-map model work better in places or conditions it has not seen before?

This project is building a careful experiment, not simply a demo. First, we measure what a model can do using only real images. Later, we can fairly compare that result with models trained using controlled synthetic changes.

Review date: August 27, 2026

What has been built

Satellite images PNG photos	Dataset loader pairs each image with its map	Model-ready data tensors and batches	Map prediction one land class per pixel
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Data exploration

- The LoveDA remote-sensing dataset was inspected before modeling started.
- The Rural training split contains 1,366 matched image-and-map pairs. The separate Rural validation split contains 992 pairs.
- Each original image is 1,024 by 1,024 pixels. Color-coded maps and overlays were checked visually to confirm the labels line up with the image.
- The maps describe seven land types: background, buildings, roads, water, barren land, forest, and agriculture. Empty or unknown pixels are ignored rather than treated as a class.

Reusable building blocks

- A reusable dataset loader turns the PNG files into the standard PyTorch format used by machine-learning models.
- A batching tool groups several image-map pairs together so training can run efficiently.
- Evaluation code calculates overall pixel accuracy, score-by-land-type, and mean Intersection over Union (mIoU). mIoU is a stricter score that rewards accurate map shapes, not just common pixels.
- Automated tests confirmed the data shapes, label handling, and metric calculations work as expected.

First real-only baseline run

A small U-Net model was prepared as the starting point. It learns from only real Rural images, with no synthetic augmentation. This is important because every later synthetic-data experiment needs a fair comparison point.

Setting	Recorded value
Training images	Train/Rural
Validation images	Val/Rural
Image size used	512 x 512
Model	UNet(base_channels=16)
Number of land classes	7
Learning method	Adam at learning rate 0.0001
Batch size and seed	1 image; seed 42
Synthetic augmentation	None

Measured results so far

Five of the planned 15 epochs were recorded. The best validation mIoU was 20.3% at epoch 3. At that point, pixel accuracy was 49.7%.

Epoch	Training loss	Validation mIoU	Pixel accuracy
1	1.668	17.7%	51.4%
2	1.374	15.4%	42.6%
3	1.228	20.3%	49.7%
4	1.145	19.4%	44.1%
5	1.086	17.3%	42.0%

How to read the early baseline

Land type	Best saved IoU
Background	39.1%
Building	12.6%
Road	0.0%
Water	24.9%
Barren	0.0%
Forest	23.8%
Agriculture	41.4%

Short baseline note

- Strongest early classes: agriculture, background, water, and forest had the highest overlap scores in the best saved epoch.
- Weakest early classes: roads and barren land were almost not recognized, while buildings were still weak. These categories are harder because they occupy fewer pixels and can look similar to nearby land.
- Did training converge? The training loss improved steadily from 1.668 to 1.086 across the five saved epochs, but validation mIoU rose and fell. That means the model was learning, but this is not yet a stable or final baseline result.
- Obvious failure pattern to investigate: thin roads and small or less common land regions are likely being missed. The validation pictures on the next page make these mistakes easier to inspect.

What happens next

- Complete a documented real-only baseline run before freezing its setup for fair comparison.
- Use the saved visualization examples to inspect where the predicted map differs from the true map.
- Only after the real-only baseline is fixed should controlled brightness, blur, noise, haze, or other synthetic transformations be tested.

Early validation examples

Each row shows the original satellite image, the known land map, the model prediction, and an error map. Red in the error map marks pixels where the prediction differs from the known map. These examples use the best saved early checkpoint from epoch 3.

